Impact of Weather and Socioeconomic Factors on Natural Gas Consumption

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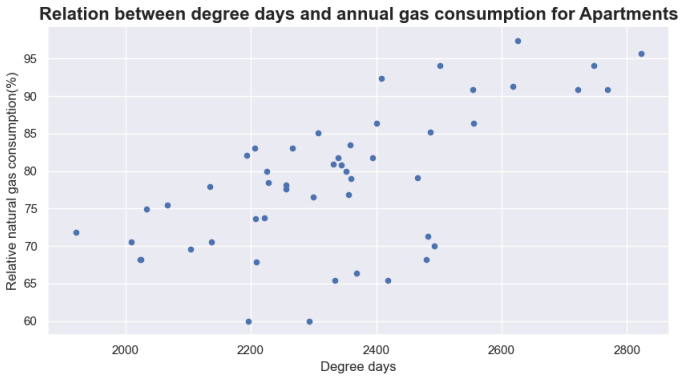
Introduction

The war in Ukraine caused an energy crisis in Europe, driving natural gas prices from €20 per MWh to over €200 per MWh, heavily affecting Dutch households, 90% of which rely on gas for heating. In 2021, households accounted for 33% of natural gas consumption in the Netherlands, leading to behavior changes like lowering thermostats and improving insulation. This study examines how degree days and average median income affect gas consumption in apartments across five Dutch municipalities—Asten, Enschede, Leidschendam-Voorburg, Leudal, and Wageningen—to inform policies promoting energy efficiency and climate goals.

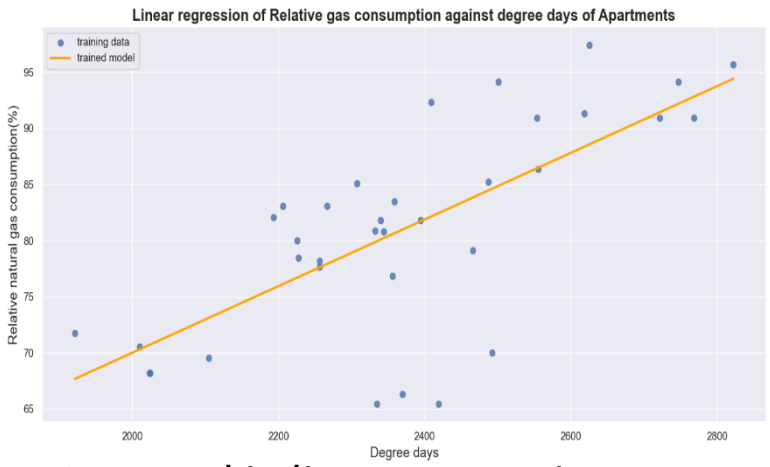
Methods

To address the question, scatter plots, line plots, and linear regression were used, covering the years 2012-2021. A scatter plot was chosen to show variability and distinct data points, revealing fluctuations. A linear plot is better for degree days, highlighting trends over time. Linear regression was selected for its simplicity and ability to quantify the relationship between degree days, average median income and gas consumption.

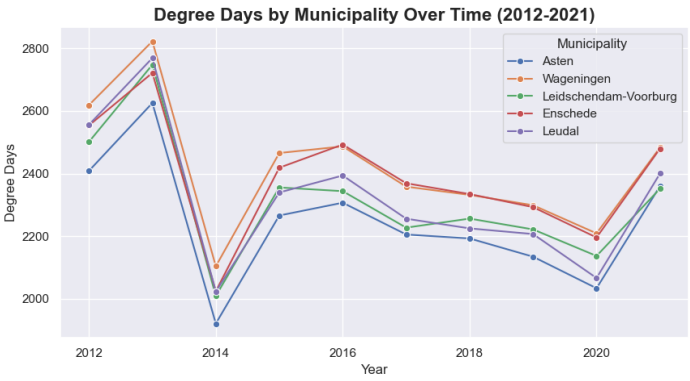
Results



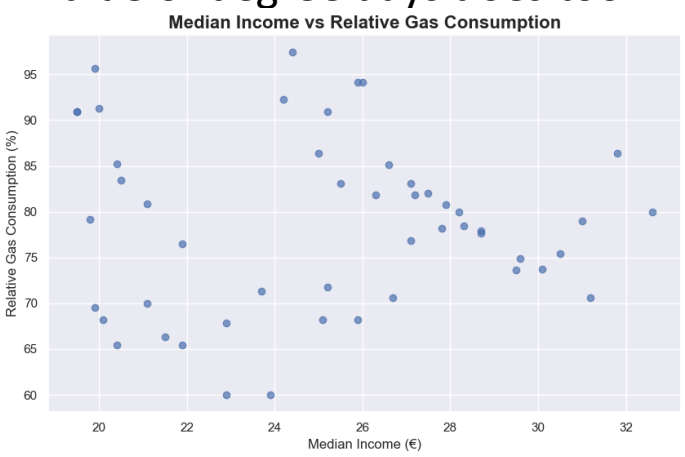
In the plot above, relative gas consumption increases when value of degree days does too.



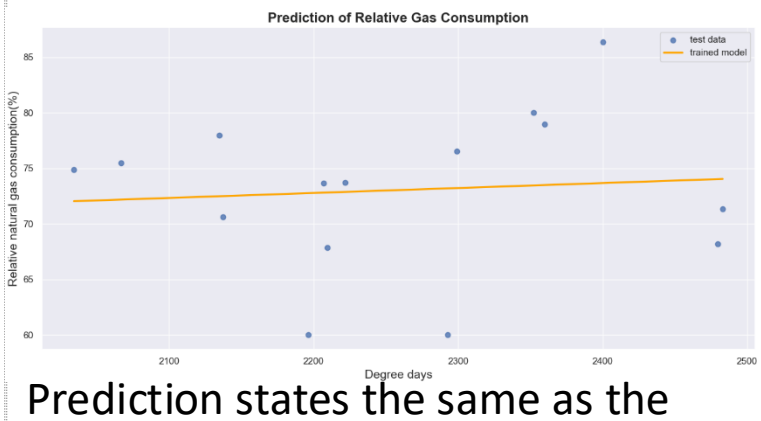
As stated in linear regression, relative gas consumption grows, when value of degree days increases.



This line plot shows the fluctuations in degree days across municipalities in the Netherlands from 2012 to 2021, highlighting seasonal temperature variations. It enables a comparison of temperature extremes, which can influence natural gas consumption and energy efficiency.



This plot, however, does not show a relation between relative gas consumption and average median income only.



Prediction states the same as the linear regression. The mean absolute error of this model is 7.5, indicating a clear relationship between the variables.

Discussion

Linear regression was applied to analyze the connection among gas usage, median income, and degree days, providing clear understanding of trends. Nonetheless, it presumes a linear connection and ignores aspects such as housing style or geographical variations. Employing non-linear regression along with the inclusion of extra variables or time series analysis might enhance the model's accuracy and enable it to identify more intricate patterns.

Conclusion

In conclusion, the analysis shows that increased degree days lead to higher gas consumption, while higher median income appears to reduce consumption, likely due to more energy-efficient housing. These insights underscore the importance of factoring in both climate conditions and socioeconomic factors when developing policies aimed at improving energy efficiency and reducing gas reliance.